

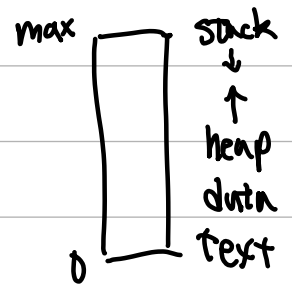
CH03 process

- process concept

실행 중인 프로그램 자체 = process

구성

- program code
- program counter, register
- stack 임시 데이터 (함수 파라미터, local variable 등, 끝나면 없어짐)
- data global, static variable
- heap dynamically allocated



- state

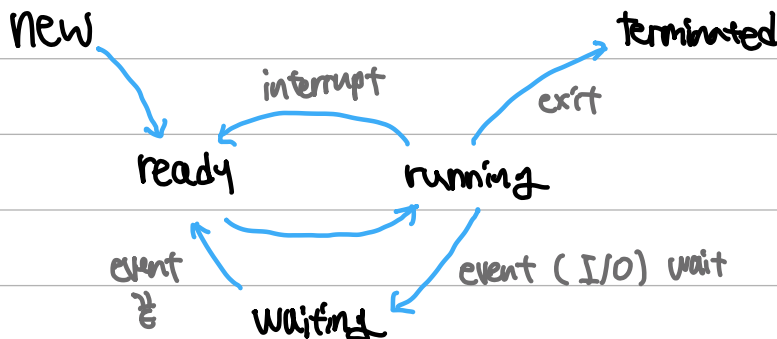
new 생성되었을 때

running CPU가 명령을 실행하고 있을 때

waiting I/O 같은 것을 대기할 때

ready CPU가 실행해줄 것 기다리고 있을 때

terminated 실행 종료



- process control block (PCB)

process state

program counter

register

priorities, scheduling queue pointers

memory management

accounting

I/O status

- 각 thread는 별도의 program counter를 가진다.

- process scheduling

scheduling queue

- job queue
- ready queue
- device queue

- short term (CPU) scheduler : 어떤 process를 실행할지 (빈번함)

long term (job) scheduler : 어떤 process를 ready queue로 보낼지 (가끔)
multi programming degree 조절

middle term scheduler: multi programming degree ↓

Ready queue 인의 process 감소

- context switch

save old process state, load saved state for new process

- interprocess communication (IPC)

[shared memory
message passing (message queue 활용) (kernel이 관리한다.)

producer-consumer problem

[unbounded buffer (queue 크기)가 무한하다고 가정)
bounded buffer

- message passing

direct communication

반로 프로세스를 명시적으로 지정

indirect passing

link만 노출되기 있음.

Synchronization

[blocking : synchronous
non-blocking ; asynchronous